# Transfer Learning for Binary Food Image Classification

INDENG 1/242B Spring 2026 Project

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# Motivation and Learning Problem

## Use case

Coarse food-image screening: classify a food image as healthy or unhealthy.

- Input feature: RGB food image $x$.
- Label: $y \in \{healthy, unhealthy\}$.
- Learned mapping: $f_\theta(x) \rightarrow y$.
- Goal: compare a task-specific CNN with transfer learning and fine-tuning.

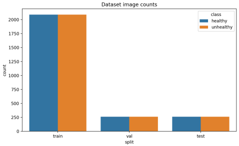
## Project questions

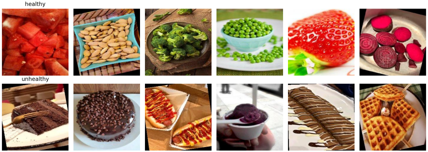
- How strong is a small CNN baseline?
- How much does ImageNet transfer help?
- Does fine-tuning improve the transferred model?
- What errors remain?

# Dataset and Preprocessing

| Split | H | U | Total |
|---|---|---|---|
| Train | 2089 | 2089 | 4178 |
| Val. | 261 | 261 | 522 |
| Test | 261 | 261 | 522 |

- 5,222 total images.
- Balanced classes in every split.
- Images resized to $128 \times 128$.
- Training: crop, flip, color jitter.
- Evaluation: resize, center crop, ImageNet normalization.



# Methodology

## Models

- Simple CNN: Conv–BatchNorm–ReLU blocks, max pooling, adaptive average pooling, dropout, linear head.
- Frozen ResNet18: ImageNet encoder fixed, binary head trained.
- Fine-tuned ResNet18: layer4-only and full-backbone variants.
- Advanced extension: supervised contrastive pretraining.

Classifier probabilities:

$$p_\theta(y = c \mid x) = \frac{\exp z_c}{\sum_{k=1}^{2} \exp z_k}.$$

Regularized objective:

$$\min_\theta \frac{1}{N} \sum_{i=1}^{N} L_{\mathrm{ce}}(f_\theta(x_i), y_i) + \lambda \|\theta_{\mathrm{train}}\|_2^2.$$

- Optimizer: AdamW.
- Regularization: weight decay, dropout, augmentation, early stopping.

# Experimental Pipeline

EDA and balanced train/validation/test split

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Simple CNN baseline

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Frozen ImageNet-pretrained ResNet18

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ResNet18 partial and full fine-tuning ablation

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Supervised contrastive branch for representation learning

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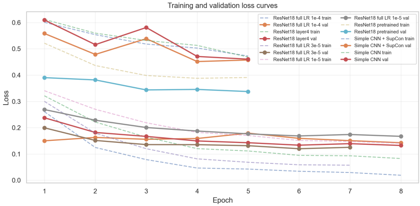
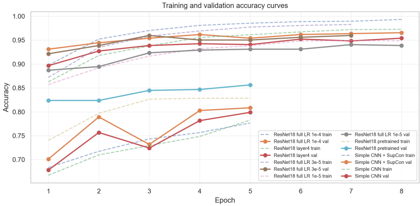
Held-out metrics, curves, confusion matrices, Grad-CAM, error analysis

The pipeline covers problem formulation, network design, objective, optimizer, regularization, advanced method, and evaluation diagnostics.

# Training and Validation Diagnostics

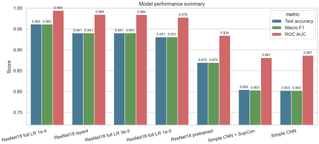



- Simple CNN variants underfit relative to the transferred ResNet18 family.
- Best fine-tuned ResNet18 keeps improving validation accuracy through the final epoch.
- A larger train–validation gap appears during fine-tuning, so validation monitoring is necessary.

# Main Results

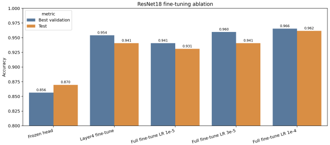
| Experiment | Val. | Test | F1 | AUC |
|---|---|---|---|---|
| ResNet18 full fine-tune, LR $10^{-4}$ | **0.9655** | **0.9617** | **0.9617** | **0.9944** |
| ResNet18 layer4 fine-tune, LR $3\times10^{-5}$ | 0.9540 | 0.9406 | 0.9406 | 0.9850 |
| ResNet18 full fine-tune, LR $3\times10^{-5}$ | 0.9598 | 0.9406 | 0.9406 | 0.9845 |
| ResNet18 frozen backbone | 0.8563 | 0.8697 | 0.8697 | 0.9344 |
| Simple CNN + SupCon fine-tune | 0.8084 | 0.8046 | 0.8033 | 0.8813 |
| Simple CNN baseline | 0.7989 | 0.8027 | 0.8027 | 0.8869 |

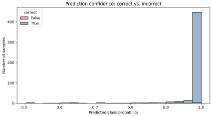



## Practical improvement

Full fine-tuning improves test accuracy from $0.8697$ to $0.9617$: a $9.20$-point absolute gain over frozen transfer.

# Ablation and Error Analysis




## Ablation finding

- Frozen head: $0.8697$ test accuracy.
- Layer4 fine-tune: $0.9406$.
- Full fine-tune at $10^{-4}$: $0.9617$.

## Residual errors

- 20 errors out of 522 test images.
- 10 healthy$\rightarrow$unhealthy and 10 unhealthy$\rightarrow$healthy.
- 12 high-confidence errors with confidence $\geq 0.80$.

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# Safety, Ethics, and Takeaways

## Deployment risks

- healthy / unhealthy labels are coarse nutritional proxies.
- Food meaning varies by portion size, ingredients, cuisine, and medical context.
- A high-accuracy model still makes high-confidence mistakes.
- User-uploaded images raise privacy and consent concerns.

## Conclusion

- Transfer learning is the dominant performance driver.
- Best model: $0.9617$ accuracy and $0.9944$ ROC AUC.
- Current system is suitable as an academic classifier, not an authoritative health tool.
- Next steps: calibration, larger backbones, external validation.

# Thank You